



Mapping intertidal topography from optical satellite time series

MAREA: the archive measures its own tide — a distributed tide gauge, a DEM with calibrated uncertainty, and its demonstrated limits

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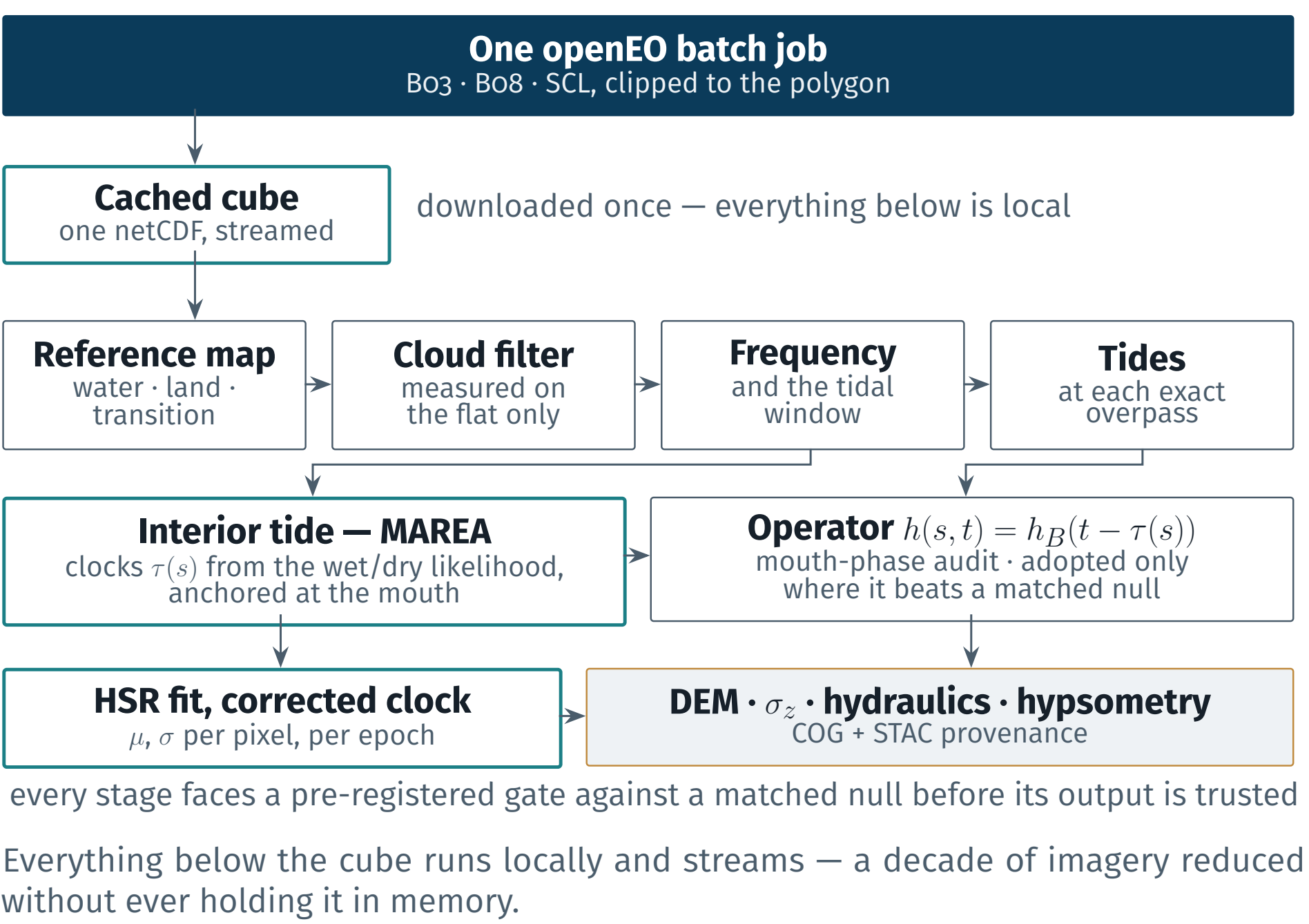
Code and data

The problem

Intertidal flats are among the most productive and fastest-changing coastal environments, yet the hardest to survey: too shallow for vessels, too soft to walk, and **never wholly visible at any single instant** — the tide hides part of the surface in every image ever taken of it. Their elevation therefore cannot be observed. It has to be *inferred*, by watching how the waterline migrates across many images taken at different tide states, and every method rests on knowing the water level at the moment each image was acquired.

This poster presents **MAREA**: the water level is no longer assumed — the archive itself measures the interior tide (a distributed tide gauge made of pixels), corrects the elevation inversion with it, and delivers the first intertidal DEM with calibrated per-pixel uncertainty, declared hydraulics, and proven limits.

One pass over the archive



Before any tide: masks, clouds, detector

The intertidal mask, from the data

Every pixel votes across the archive: permanent water, permanent land, or **transition**. The transition class, filtered by its water frequency against the tidal window, is the intertidal — **31,315** pixels in Villaviciosa, no chart or DEM consulted.

Clouds measured on the flat only

A cloud three kilometres inland does not obscure the flat, so cloudiness is scored *inside the dilated transition zone alone*, per pixel. Only 141 of 1379 scenes are fully clear — but **415** have half the flat visible, and every clear pixel in them enters the fit.

The water detector, chosen by experiment

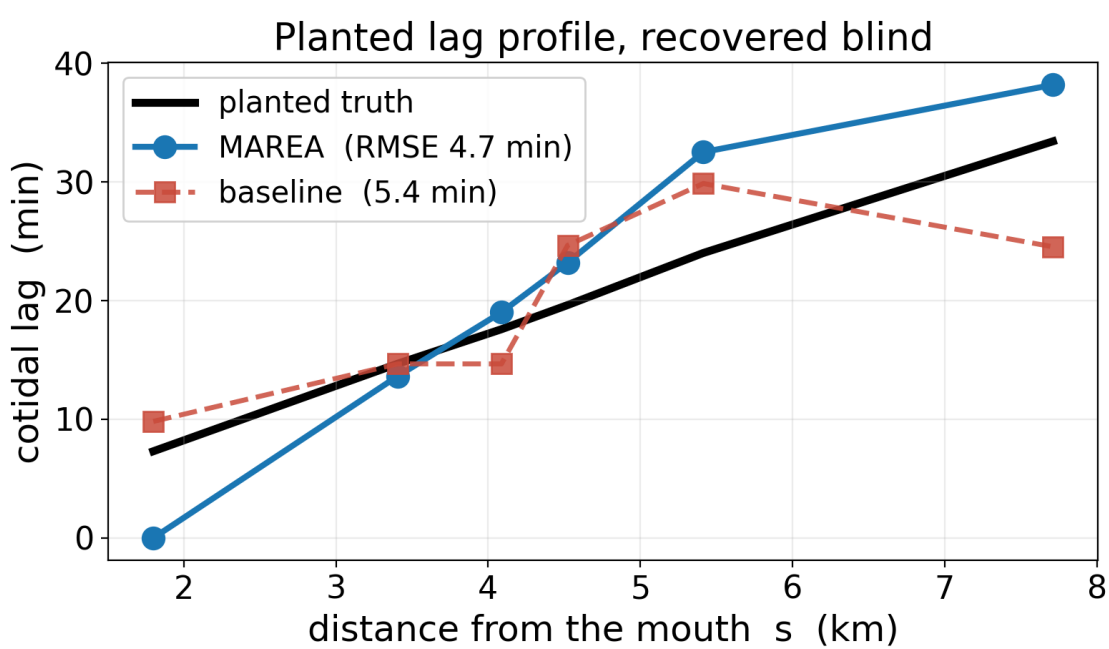
Four indices ran over byte-identical dates and masks, ranked by correlation of water frequency with LiDAR elevation (a lower pixel must flood more often). **NDWI** wins ($\rho = -0.624$): both bands are native 10 m, so no resampling smears the waterline that carries the signal.

MAREA: the archive measures its own tide

Every published method assumes **one uniform water level per scene**, from an ocean model at the mouth. Inside an estuary the tide arrives late and deformed — minutes of clock error become tens of centimetres of mis-assigned level. MAREA (*Mudflat Altimetry via Remotely-Estimated tides from the Archive*) measures that interior tide from the imagery itself.

- Each pixel is a binary threshold sensor: wet when the water tops its elevation. Its wet/dry sequence over 465 dated scenes records *when* the water crossed it.
- Thousands of sensors share the same water. Band the estuary by geodesic distance s from the mouth and ask each band: *which clock makes your wet/dry record most probable?*
- Score a candidate lag τ by the profiled Bernoulli likelihood — every pixel granted its best elevation and width, so only the clocks remain as unknowns, anchored to $\tau = 0$ at the mouth:

$$\mathcal{L}(\tau) = \sum_p \max_{z_p, \sigma_p} \sum_t \log \left[P_{pt}^{y_{pt}} (1 - P_{pt})^{1-y_{pt}} \right], \quad P_{pt} = \Phi \left(\frac{h_B(t-\tau) - z_p}{\sigma_p} \right)$$



The identifiability gate: a lag profile planted in a calibrated digital twin is recovered blind with **4.7 min** RMSE, beating the waterline-timing baseline from the literature (5.4 min).

What is measurable — and the operator that respects it

The binary likelihood is exactly invariant under $(\alpha, z, \sigma) \rightarrow (c\alpha, cz, c\sigma)$: a theorem, not a limit of effort. It dictates the division of labour:

Quantity	From imagery?	Anchor
Phase lag $\tau(s)$	Yes — ± 5 min	mouth band, $\tau = 0$
Boundary phase error	Yes — ± 4 min	mouth pixels audit it
Amplitude / gain	No (theorem)	ocean-model ensemble
Vertical datum	No (theorem)	ocean model

A planted +12 min error of the boundary model masquerades as **up to +21 min** of fictitious “estuarine transfer” — the mouth audit removes it before it can. The final operator $h(s, t) = h_B(t - \tau(s))$ carries a lag *only* where it beats a matched null (the archive re-simulated with a uniform tide: same pixels, same clouds, same noise), and passed its own gate: planted damage recovered, no harm at the mouth, and a fixed point of its own estimation. Never calibrated with tide gauges — gauges only *validate* where they exist.

Villaviciosa, measured blind: on the model’s clock up to 5.4 km; the head of the ria runs **+26 min late** — 3 of 4 independent estimators agree, outside the null.

HSR: elevation from the whole wetting curve

A 10 m pixel on the waterline is not wet or dry — it is **partially flooded**, and its NDWI is the linear mixture of its wet and dry fractions. As the tide rises that fraction follows the pixel’s own internal hypsometry, so its NDWI against tide height is a sigmoid whose parameters are physical, not fitted conveniences:

$$\text{NDWI}_p(t) = a_p + b_p \Phi \left(\frac{h_t - \mu_p}{\sigma_p} \right)$$

μ_p — median pixel elevation

Read from the *entire* curve rather than from one threshold crossing, so every clear observation contributes and the estimate is sub-decimetre.

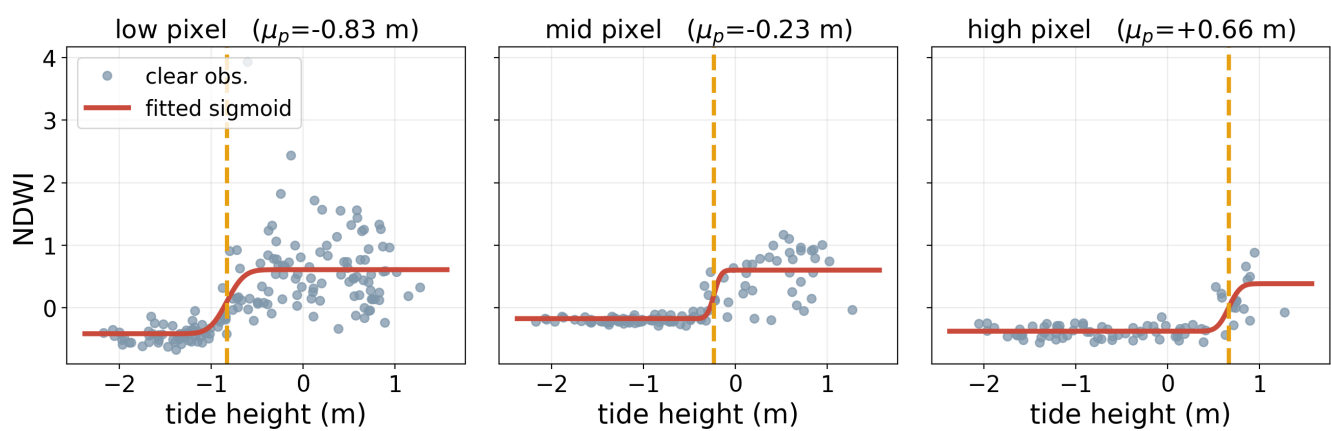
σ_p — sub-pixel relief

The roughness *inside* the 10 m cell: a product no threshold-crossing method can produce, because it discards exactly the mixed pixels that carry it.

Fit quality — a Cramér–Rao bound per pixel

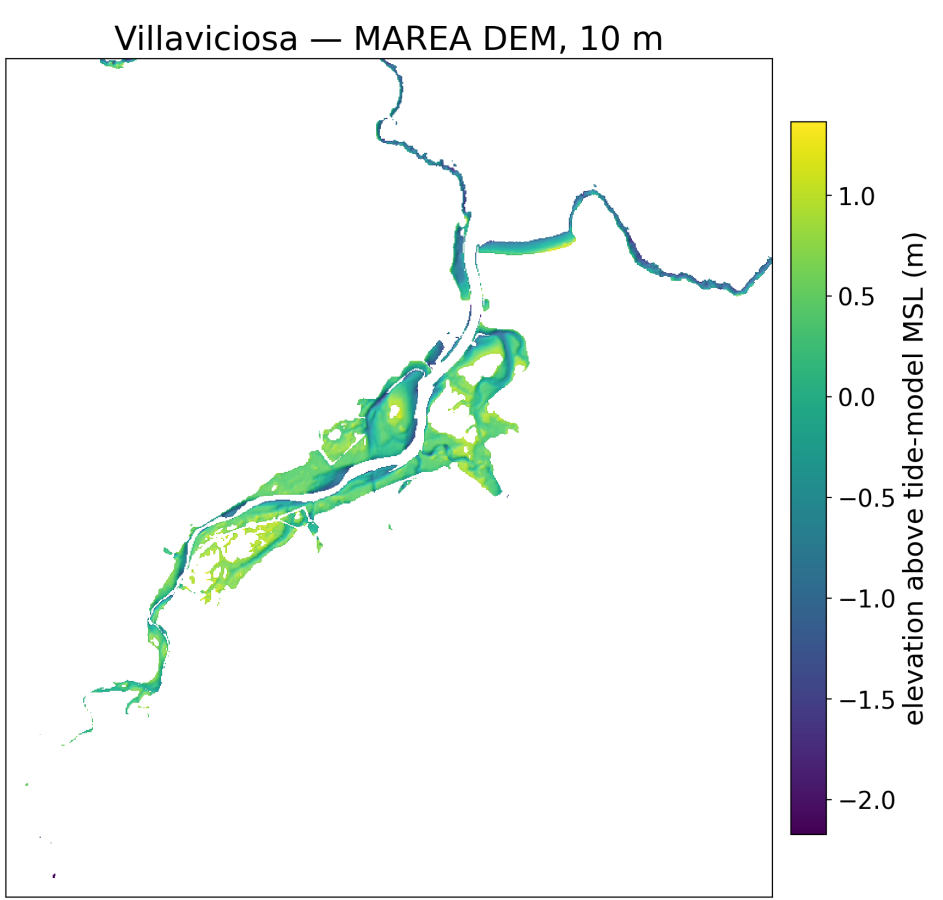
An uncertainty that is derived rather than assumed, and that doubles as automatic quality control.

The amplitude terms (a_p, b_p) have a closed-form solution per candidate (μ, σ) : the fit is a cheap grid sweep, tractable over millions of pixels. With MAREA, h_t is each band’s **corrected clock**.



Three real pixels, low to high on the flat. The sigmoid is not imposed on the data — it is what mixed pixels do as the tide crosses them, and its centre μ_p is the elevation.

A DEM that knows what it knows



Villaviciosa, MAREA DEM at 10 m (operator-corrected clock; declared interpolation flags; 2.6 % unmeasurable pixels stay honest NaN).

Three layers no published intertidal DEM carries:

σ_z per pixel, calibrated

Tabulated from planted recoveries in the calibrated digital twin, by tidal headroom and observation count. Verified conservative: the 68 % interval covers **83 %** of real errors — never optimistic.

σ_{topo} vs. level decomposition

The fitted transition width carries the tide-model error convolved in; de-convolved, median sub-pixel relief is **0.177 m** (35 % of the width’s variance was water level, not ground).

Declared hydraulics

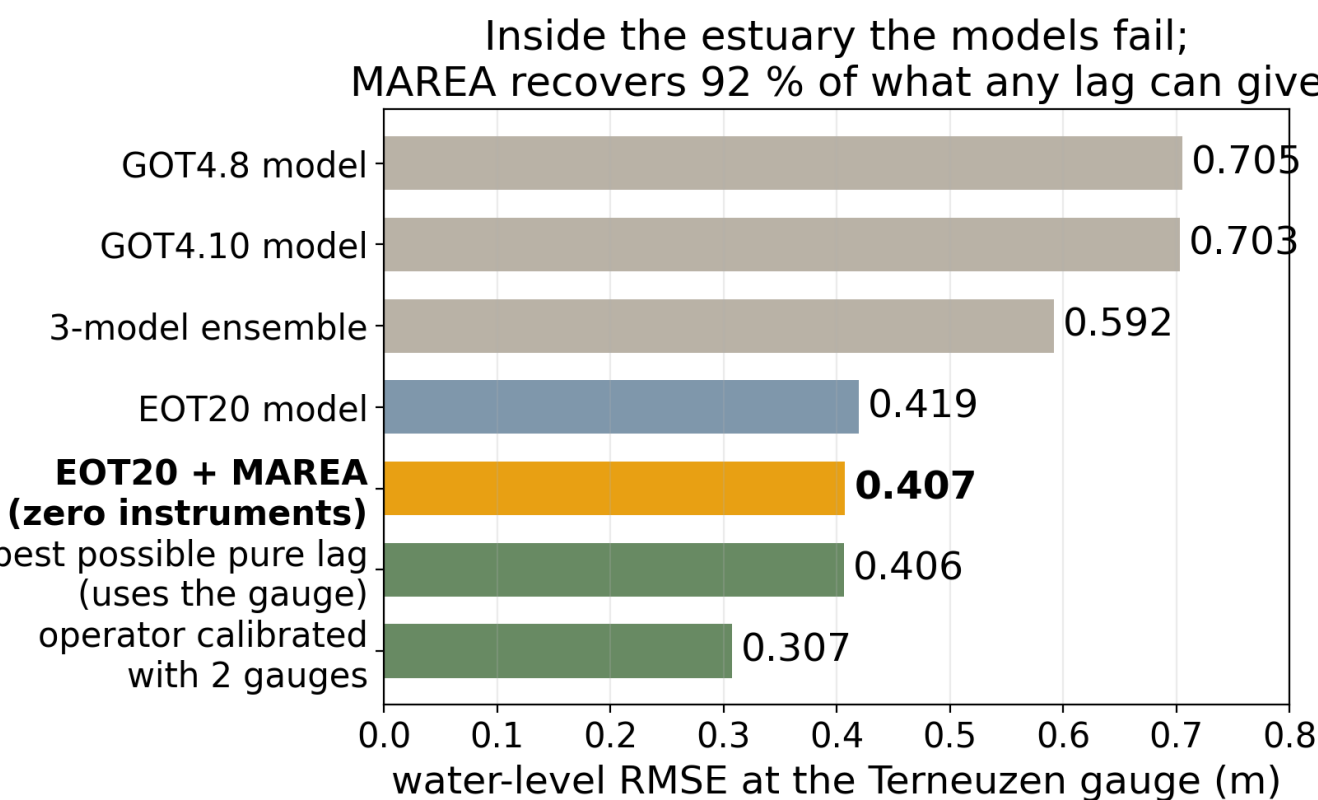
Depression filling yields each pixel’s spill elevation and ponding depth: **14 %** of the intertidal sits behind a barrier (median 1.2 m). The flag is validated against the real ponding signature, its bounded recall printed on the product.

External validation: never worse, better where it matters

Elevation against independent surveys of the flats themselves, and water level against gauges never used in calibration. Centred RMSE, metres:

Site (truth)	Regime	uniform	MAREA
Tagus interior (EMODnet, 286k px)	estuary, $\tau \approx 45'$	0.279	0.247
Aveiro lagoon (LiDAR, 333k px)	branched, τ : 41–56'	0.590	0.580
Vadehavet (EMODnet, 344k px)	open flat	0.275	0.275*
Villaviciosa (own RTK, 217 pts)	short deep ria	0.211	0.211*

*the operator *declines* where the lag does not beat the null or does not help — equality is the correct verdict there, verified, not assumed. In Aveiro the gain concentrates where the lag grows: +24 to +53 mm per band.



The Scheldt, blind: MAREA’s lag gradient (**0.72 min/km**) matches the tide gauges’ 0.9, and at the Terneuzen gauge — never used in calibration — its level sits 1 cm from the best any pure lag could do.

Three demonstrated limits (negative results we keep)

Every claim faced a pre-registered gate against a matched null; three died with a clean control, and each is a citable boundary for the field:

Gain is invisible to binary data

Twice (interior gain, boundary gain): the likelihood cannot separate amplification from a rescaled seabed. The pipeline *exposes* the flat likelihood instead of printing a number.

Per-limb clocks are not adoptable at 465 scenes

Three versions, three failure modes caught (likelihood runaway, multiple-comparison adoption, insufficient power under the null threshold). Ponding stays a simulator mechanism and a quality flag — not a correction. Reopens with the 10-year archive.

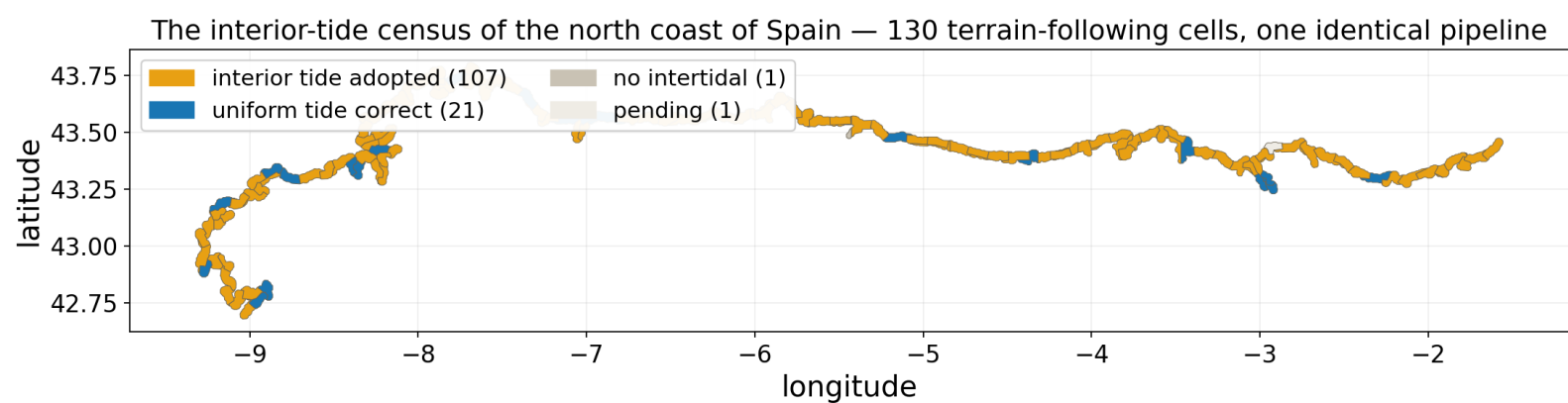
Censoring hides itself

A fully censored pixel refits at its spill level, erasing the geometric evidence of its own pit: any detector built from the refit map is recall-bounded (≈ 0.4 –0.5, measured).

Point-vs-pixel: comparing GNSS points to 10 m pixel medians adds $\sigma \approx 0.21$ m that belongs to the *yardstick* — the “slope compression” this literature reports is largely that artefact (true $c = 1.04$). The method’s own error is ≈ 0.13 –0.15 m.

The north coast: 130 cells, one census

Nothing is site-specific: every cell runs the identical pipeline — same configs, same seeds, zero per-site tuning. Terrain-following cells from the real coastline, cubes queued server-side on openEO, harvested and processed in a resumable worker pool; one tide-model call per cell.



The first **interior-tide census** of the coast — decided by the method itself: **107** of 128 mapped cells carry a measured interior tide; 285 km² of intertidal, 2.85 M pixels with elevation.

Reproducibility

Every product is a cloud-optimised GeoTIFF with a STAC item beside it carrying the run’s resolved parameters, so provenance travels with the file rather than living in a notebook. The toolkit is deliberately granular: no `run_everything()`, one function per scientific step.

Bishop-Taylor, R. et al. (2019). Between the tides: modelling the elevation of Australia’s exposed intertidal zone at continental scale. *Est. Coast. Shelf Sci.* 223, 115–128.

Hart-Davis, M. et al. (2021). EOT2o: a global ocean tide model from multi-mission satellite altimetry. *ESSD* 13, 3869–3884.

Schramm, M. et al. (2021). The openEO API. *ISPRS Int. J. Geo-Inf.* 10(3), 125.

Granadeiro, J. P. et al. (2021). Using Sentinel-2 images to estimate topography, tidal-stage lags and exposure periods over large intertidal areas. *Remote Sens.* 13(2), 320.

Contains modified Copernicus Sentinel-2 data (2016–2025) via the Copernicus Data Space openEO backend. Tides: EOT2o (SEANOE), GOT4.10 (NASA GSFC) via pyTMD. Field validation: RTK GNSS campaign, Ria de Villaviciosa.